```python
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
Created on Tue Feb 24 00:26:52 2026

@author: frankie
Description:
Enter a principle amount of a CD and year to maturity of CD. Determine the
interest rate based on the amount of the principle and maturity (see below).
Calculate first year interest (principle x interest rate). Display principle, interest
rate and the interest amount for first year.
Principle                Years to Maturity   Interest Rate
>$100,000                5                   6%
$50,000 to $100,000      10                  5%
$50,000 to $100,000      5                   4%
Any other principle and years                2%
"""

# Enter information by user
principle = float(input("Enter the principle amount".ljust(30)+": "))
yearToMaturity = int(input("Enter the year to maturity".ljust(30)+": "))

# Checking interest rate

if principle > 100000 and yearToMaturity == 5 :
    interestRate = 0.06
elif principle >= 50000 and principle <= 100000 and yearToMaturity == 10 :
    interestRate = 0.05
elif principle >= 50000 and principle <= 100000 and yearToMaturity == 5 :
    interestRate = 0.04
else:
    interestRate = 0.02

# Calculate interest amount
interestAmountForFirstYear = principle * interestRate

# Display the output

print("".ljust(53,"="))
print("Principle".ljust(30),":",f"{principle:20,.2f}")
print("Interest rate".ljust(30),":",f"{interestRate:20,.0%}")
print("Interest amount for first year".ljust(30),":",f"{interestAmountForFirstYear:20,.2f}")
```

| | |
|---|---|
| M6P1_Result.pdf | 2/25/26 2:31 PM |
| M6P1.py | 2/24/26 12:10 AM |
| M6P2_Result.pdf | 2/25/26 2:33 PM |
| M6P2.py | 2/24/26 12:26 AM |
| M6P3.py | 2/24/26 12:44 AM |
| M6P4.py | 2/24/26 12:53 AM |
| M6P5.py | 2/24/26 1:02 AM |

Help | Variable Explorer | Debugger | Profiler | Plots | Files

Console 1/A

```
In [34]: %runfile '/Users/frankie/Dropbox/Harper College/CIS
2026/M6/M6P3.py' --wdir
Enter the principle amount    : 150000
Enter the year to maturity    : 5
=====================================================
Principle                     :           150,000.00
Interest rate                 :                   6%
Interest amount for first year :            9,000.00

In [35]: %runfile '/Users/frankie/Dropbox/Harper College/CIS
2026/M6/M6P3.py' --wdir
Enter the principle amount    : 100000
Enter the year to maturity    : 10
=====================================================
Principle                     :           100,000.00
Interest rate                 :                   5%
Interest amount for first year :            5,000.00

In [36]: %runfile '/Users/frankie/Dropbox/Harper College/CIS
2026/M6/M6P3.py' --wdir
Enter the principle amount    : 100000
Enter the year to maturity    : 5
=====================================================
Principle                     :           100,000.00
Interest rate                 :                   4%
Interest amount for first year :            4,000.00

In [37]: %runfile '/Users/frankie/Dropbox/Harper College/CIS
2026/M6/M6P3.py' --wdir
Enter the principle amount    : 150000
Enter the year to maturity    : 10
=====================================================
Principle                     :           150,000.00
Interest rate                 :                   2%
Interest amount for first year :            3,000.00

In [38]:
```

IPython Console | History

Update available | Inline | Conda: spyder-runtime (Python 3.12.11) | LSP: Python | Line 39, Col 1 | UTF-8 | LF | RW | Mem 48%